Lecture 9

Tuesday, 19 September 2023 10:35 AM

Temperature-programmed Techniques

TP

- R → Reduction
- O → Oxidation
- S → sulfidation
- D → Desorption
- SR → surface reaction
- RS → reaction spectroscopy

Intent behind the experiment

- track reduction/oxidation state of some part of solid → and then characterize the products given out
- specific use of H_2S to reduce
- displacement or temperature control

Info gained

+ Desorption vs. Adsorption energy : strength of binding (physisorption vs. chemisorption)
+ TPSR → rates of reaction & steps ; types/barriers of reaction
+ State of catalyst site (ion vs. metal state)
+ Quantify no. of sites (via TPD)

Multiple active sites ? → powdered, porous material complicates TPR

↳ Temp. peak does not always correspond to reaction rate of active site

TCD → thermal Conductivity detector (b/w reference & analyte stream)

[Wheatstone bridge]

↳ Gas composition affects the resistor resistance (due to change in thermal conductivity)
↳ changing current
↳ b/w reference & analyte

MS → sorting ions based on mass/charge ratio

Ionization

- electron impact → Large molecular fragments
- Chemical Ionization (soft ionization) → electron transfer through collision of molecules

Parention → fragments due to cleavage → major ion & then H